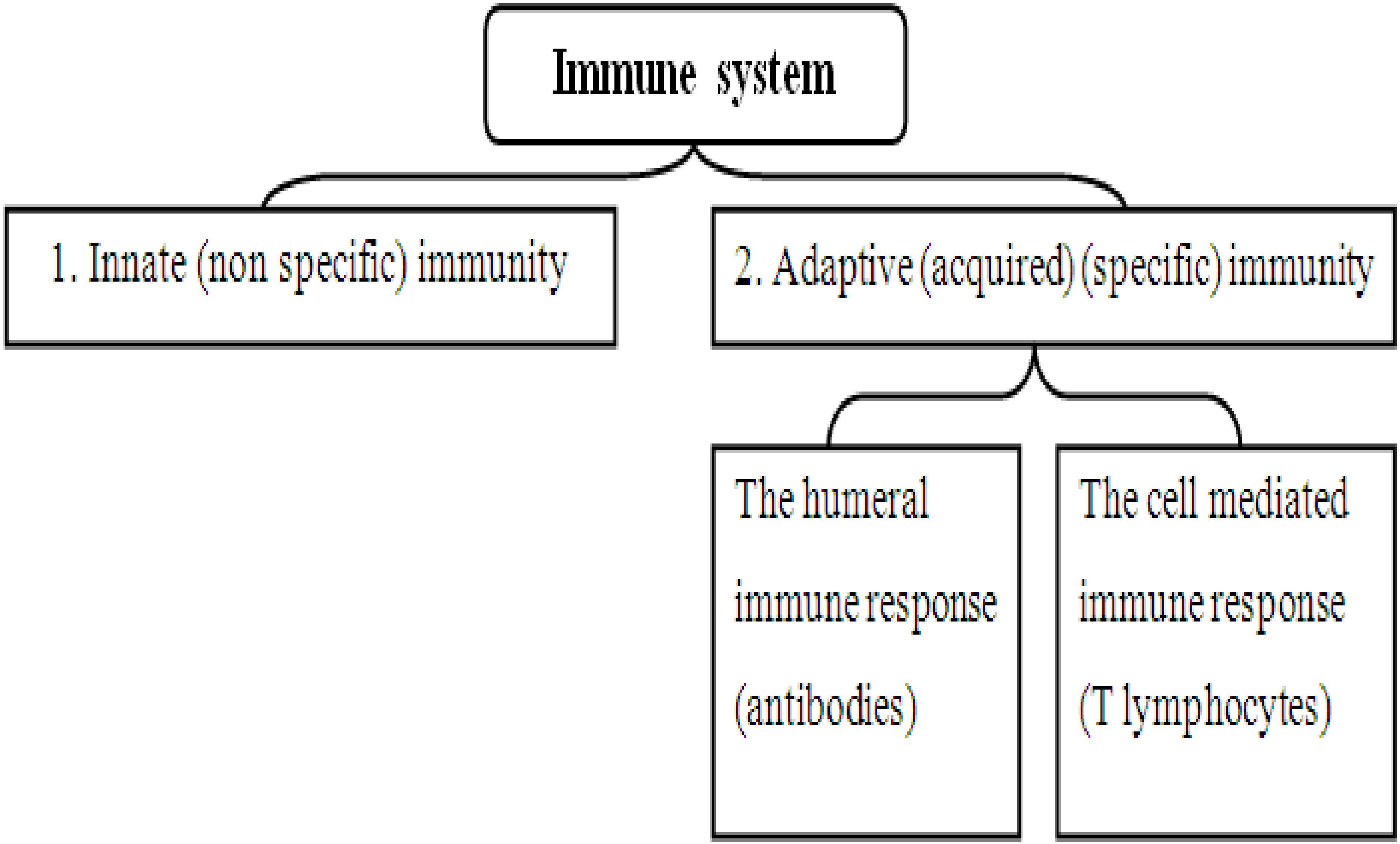
Immune system 1


Immune system
1. Innate (non specific) immunity
2. Adaptive (acquired) (specific) immunity
The humeral immune response (antibodies)
The cell mediated immune response (T lymphocytes)

1. Innate immunity

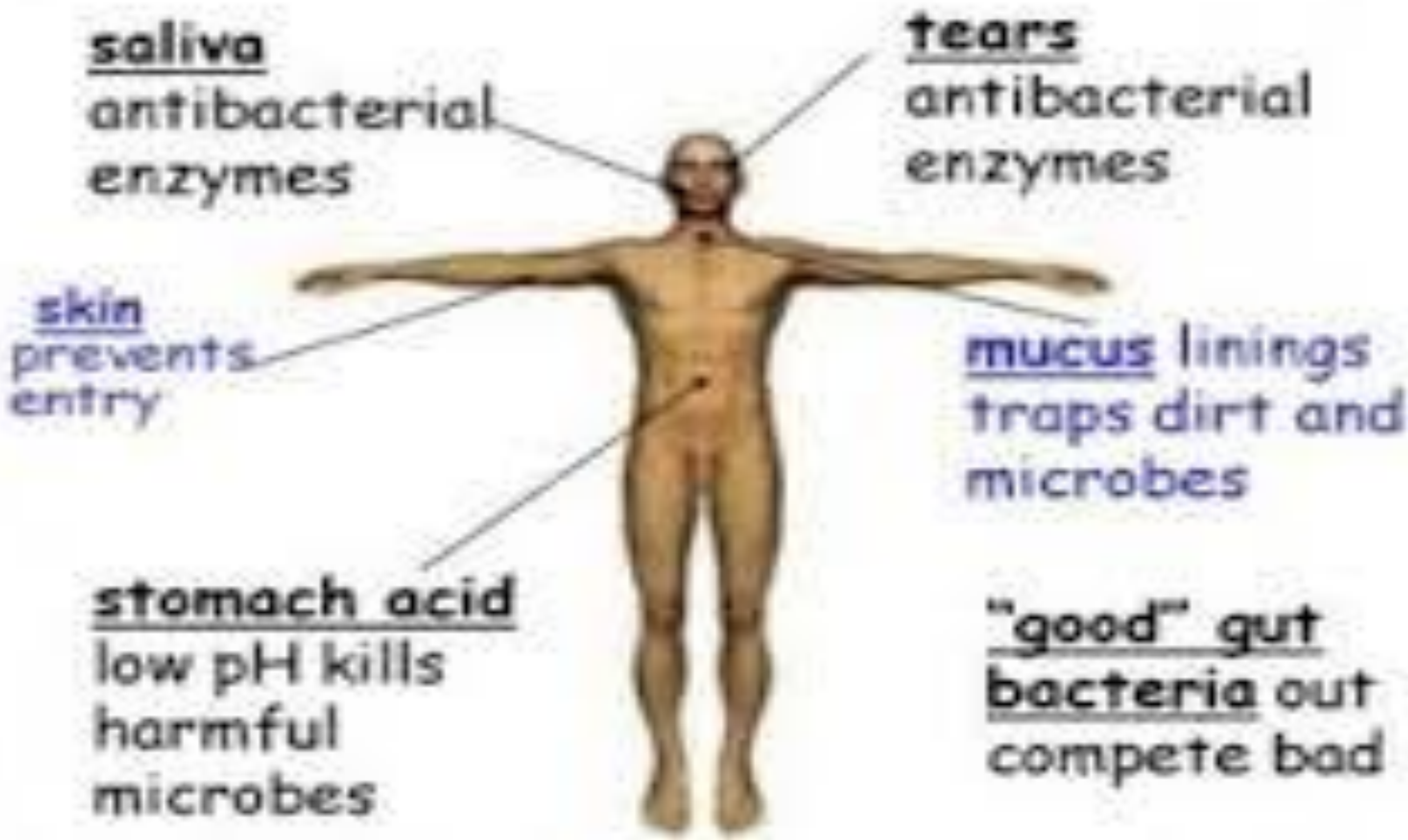
- **Characters:**

1. Natural inborn barriers.
2. Not acquired through previous exposure to infectious agents.
3. The first line of defense.
4. Non-specific against any foreign invader.
5. Rapid in action but short-lived.

- **Consists mainly of:**

1. **Mechanical barriers** e.g. intact skin and mucus membranes.
2. **Chemical barriers** e.g. HCL of the stomach.
3. **Normal bacterial flora**: They suppress the growth of pathogenic bacteria.

First Lines of Defence

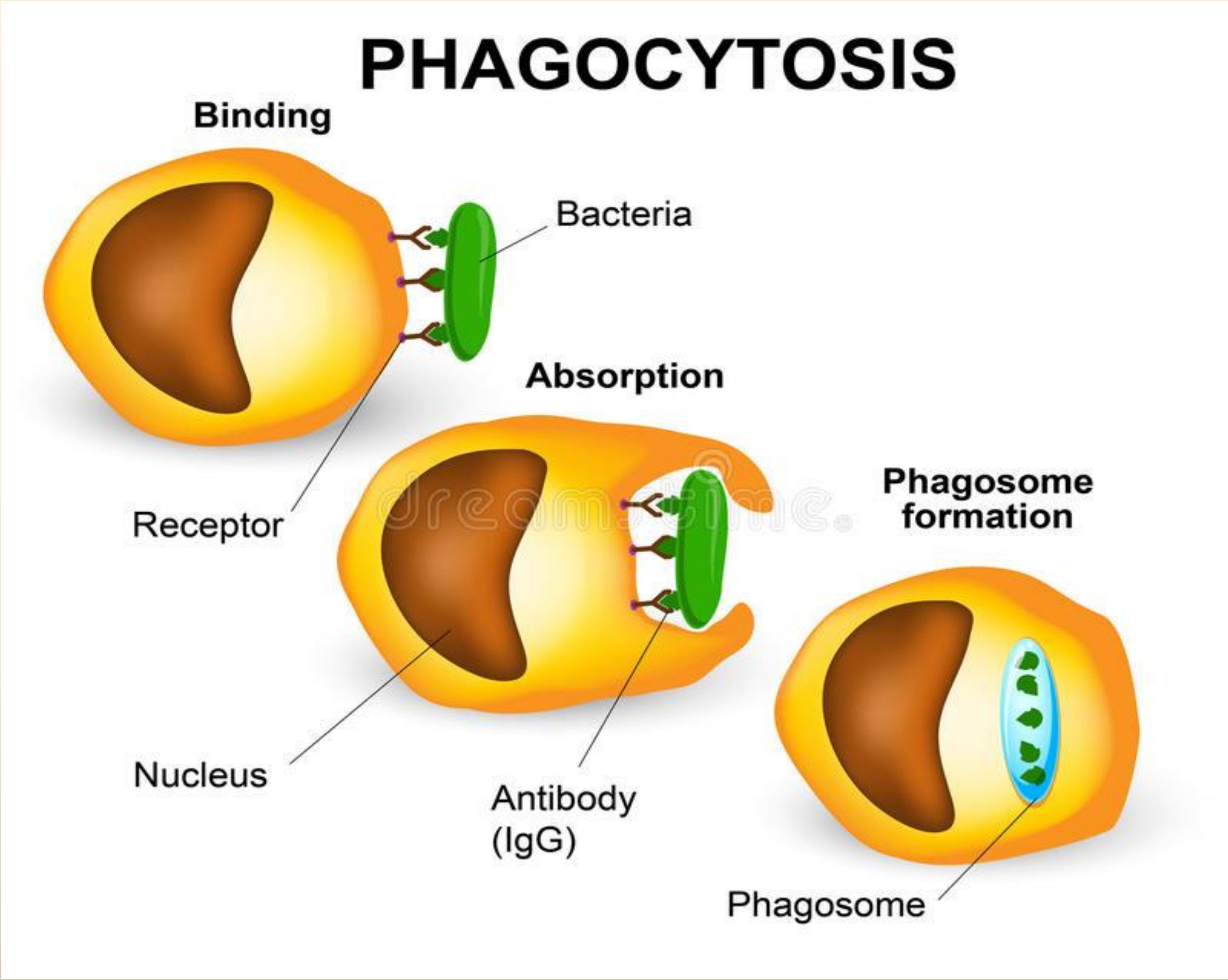


4. Circulating proteins that mediate destruction of microorganisms e.g. acute phase proteins.

5. Cells of innate immunity:

a. Phagocytes → phagocytosis.

b. Natural killer (NK) cells → kill the target cells.


PHAGOCYTOSIS
Binding
Bacteria
Absorption
Phagosome
formation
Receptor
Nucleus
Antibody
(IgG)
Phagosome

2. Adaptive immunity

- **Characters:**

1. Directed against a single organism (specific).
2. Acquired due to previous exposure to the same organism as in infection or vaccination.

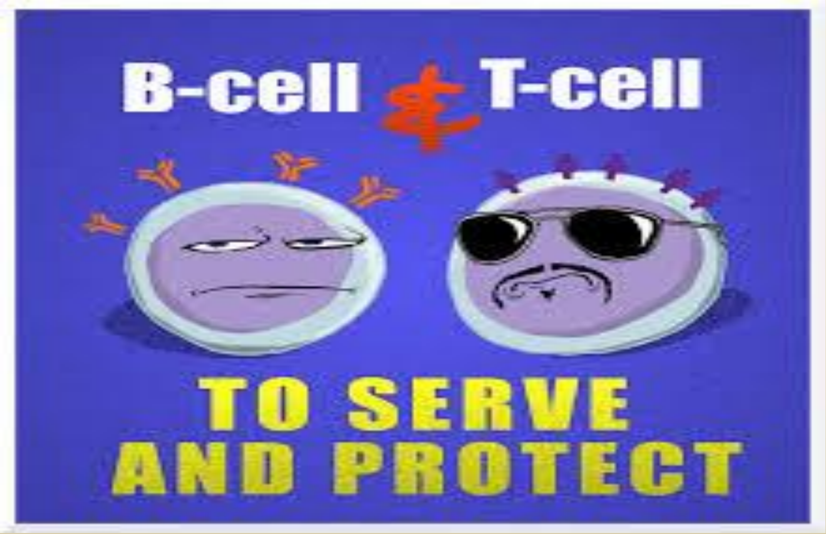
3. Depends on lymphocytes which have specific receptors.

4. Have memory.

- **<u>Cells of adaptive immunity:</u>**

a. B lymphocytes.

b. T lymphocytes (T Helper and T Cytotoxic).



	Natural	**Adaptive**
Cells	Phagocytes. Natural killer cells	B Lymphocytes. T lymphocytes (Helper & cytotoxic).
Molecules	Acute phase proteins	Antibodies
Speed & duration	Rapid, short lived	Slow, prolonged.
Development of memory	No	Yes

Induction of the immune response

❖ **Stage I:**

- Antigens (foreign substances) pass through lymphatics to the draining lymph nodes.

❖ **Stage II**:

- In the draining lymph node, the antigen attaches to its receptors on the lymphocyte (specificity) lymphocytes will proliferate.

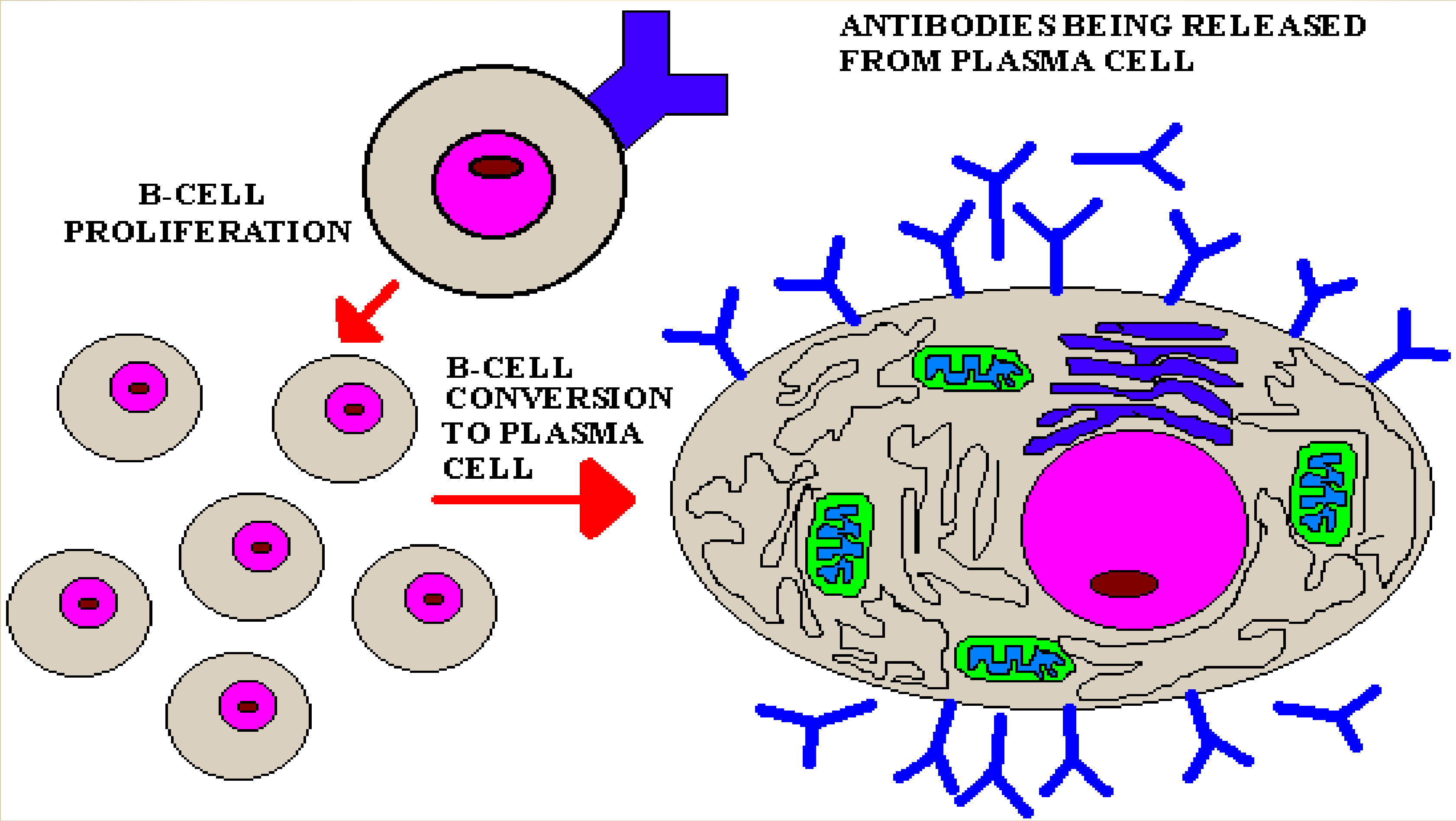
❖ **Stage III:**

- The lymphocytes leave the lymph node and travel to the site of infection by blood to fight against infection.

Mechanism of action of the specific immune response

I.Humeral immunity:

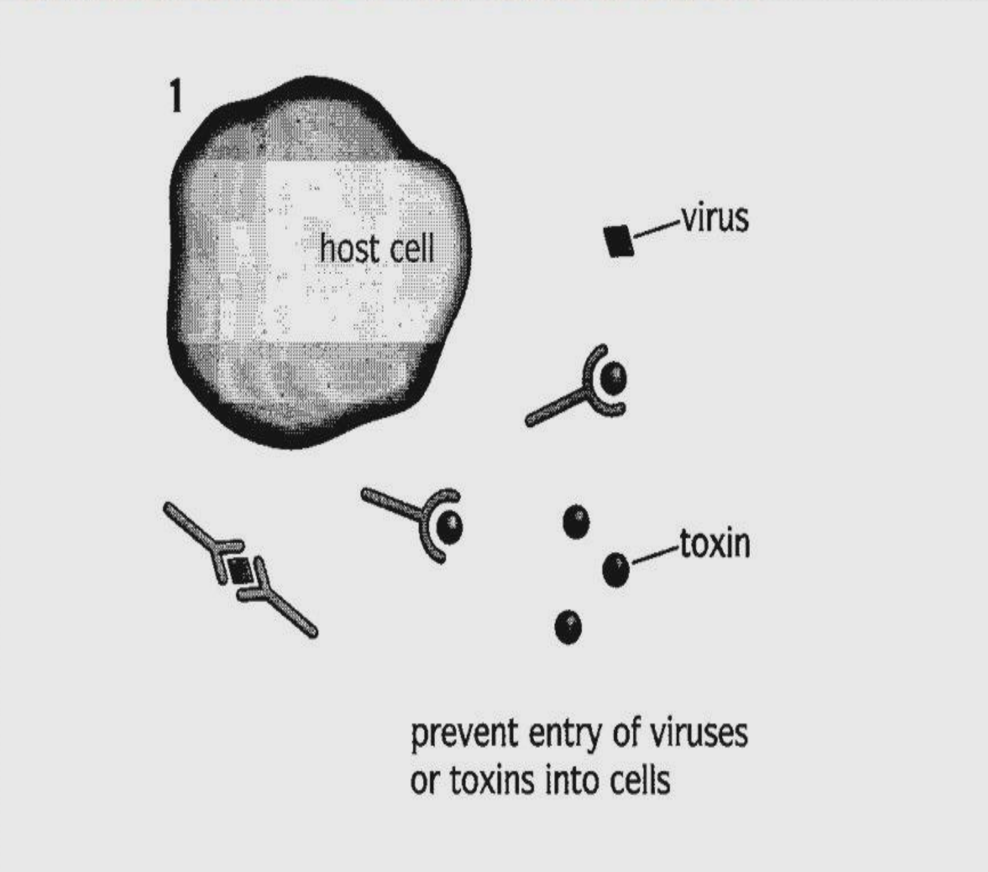
- **B lymphocytes** → Activation → proliferation → plasma cells → **secret antibodies**.

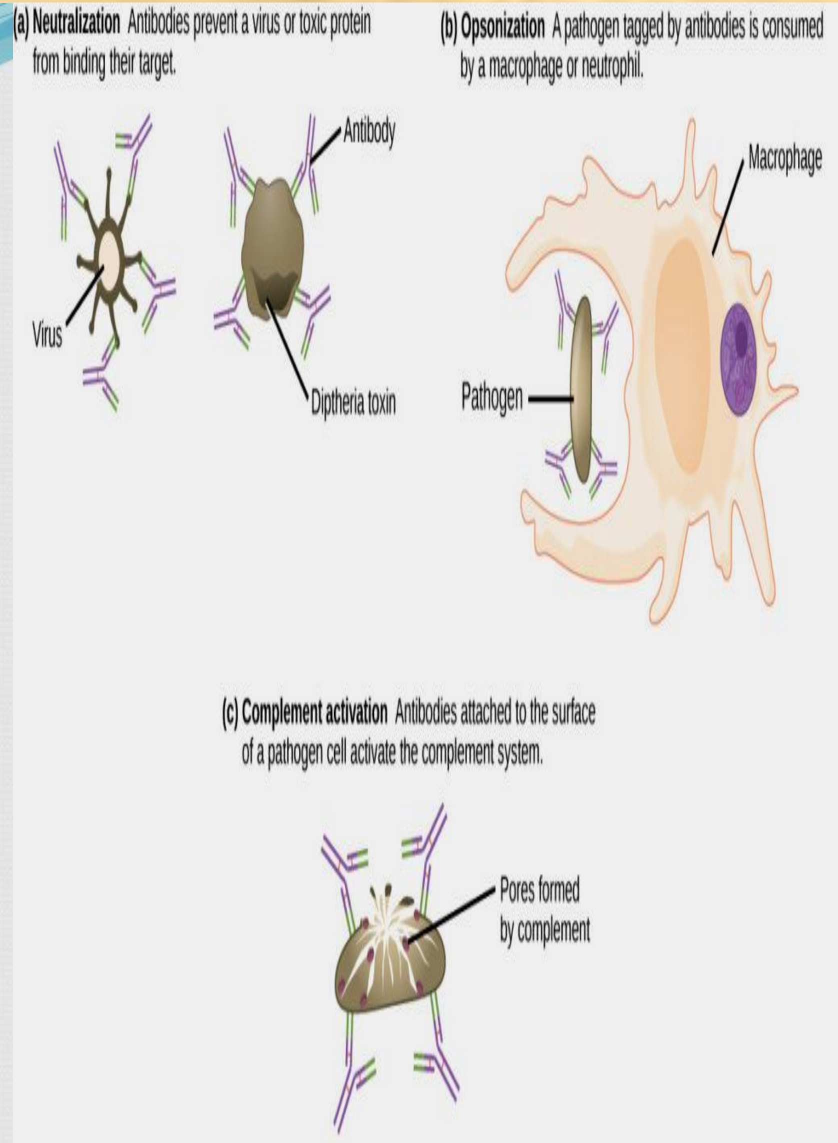

ANTIBODIES BEING RELEASED FROM PLASMA CELL
B-CELL PROLIFERATION
B-CELL CONVERSION TO PLASMA CELL

- **Functions of antibodies:**

1. Neutralizes bacterial toxins.
2. Antibodies cover the virus and prevent the attaching to cells.
3. Antibody molecule combines with more than one antigen → clumping → easily phagocytosed.

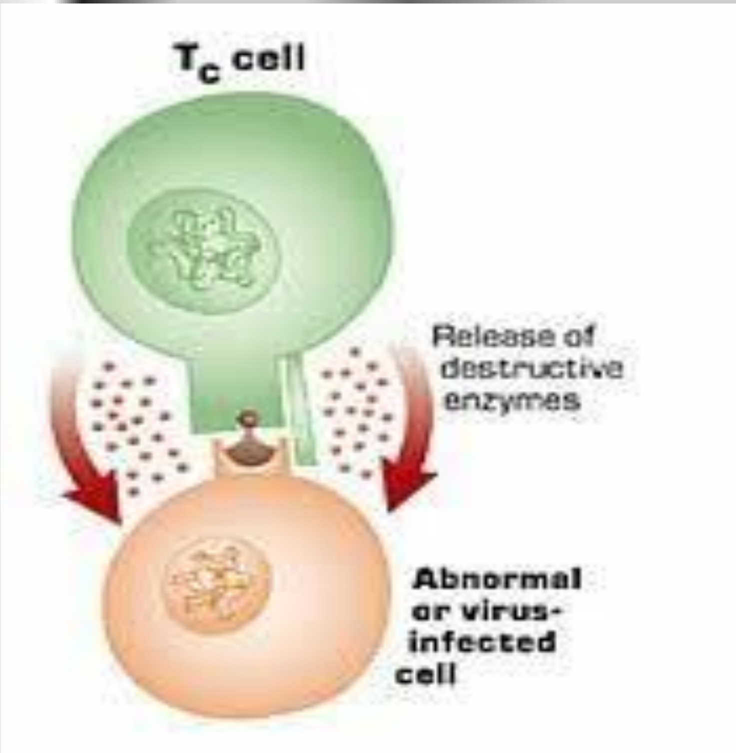
Functions of antibodies

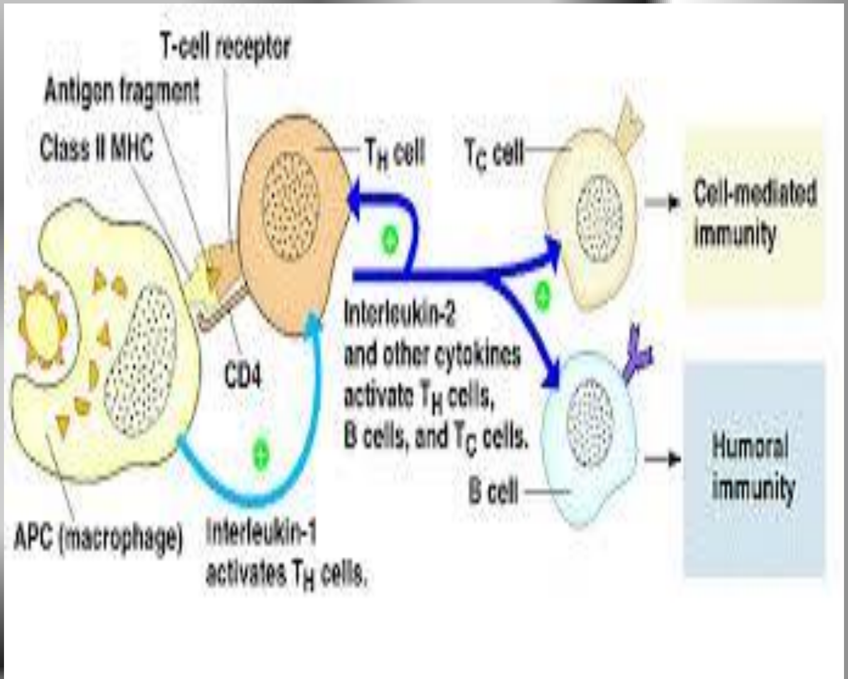




II. Cell mediated immunity

- **T helper** → Activation → proliferation → effector cells → **secret cytokines**.
- **T cytotoxic** → Activation → proliferation → effector cells → **kill target cell**.


T_C cell
Release of destructive enzymes
Abnormal or virus-infected cell


T-cell receptor
Antigen fragment
Class II MHC
T_H cell
T_C cell
Cell-mediated immunity
CD4
Interleukin-2 and other cytokines activate T_H cells, B cells, and T_C cells.
B cell
Humoral immunity
APC (macrophage)
Interleukin-1 activates T_H cells.

Questions ???

- Q1:compare between natural and adaptive immunity??
- Q2: What are the functions of antibodies?
- Q3 Mention 3 characters of adaptive immunity?
- Q4 Mention 3 characters of innate immunity?

- MCQ

1-Which of the following cell secrets antibodies ……………

(T helper ,T cytotoxic , Plasma cell)

2-Which of the following cell can kill tumour cell directly…………….

(T helper ,T cytotoxic , Plasma cell)

3-Which of the following is innate immunity cell……………..

(T cytotoxic , Plasma cell, phagocytes)

4- T helper is the orchestra of immune system as it…………….

(serete cytokines, phagocytose antigen , direct cell killing)

5-Specific immunity characterized by ……………..response

(slow, rapid , intermediate)

True or false:

1- one of mediated immunity cells is Natural killer cell ( )

2-Innate immunity is directed against a single organism ( )

3- Innate immunity characterized by having memory after infection ( )

4-T &B lymphocytes responsible for production of innate immunity ( )

5-T helper considered as the orchestra of the immune system as it secrete different cytokines ( )

6- antibodies cause direct lysis of the organism ( )